

Multiple Choice (10 marks)

1. Find the GCD of 25 and 55.
 - (a) 275
 - (b) 15
 - (c) 20
 - (d) 50
 - (e) 11
2. Compute the area of the triangle whose altitudes have lengths 10, 12, and 15.
 - (a) $56\sqrt{15}$
 - (b) $60\sqrt{7}$
 - (c) $240\sqrt{7}/7$
 - (d) $48\sqrt{15}$
 - (e) $120\sqrt{15}$
3. A number x equals $2^{15} \cdot 3^6$. What number's cube equals x ?
 - (a) 125
 - (b) 343
 - (c) 64
 - (d) 216
 - (e) 512
4. The American flag is usually made with its width and length in the ratio of 10 to 19. Which dimensions are in the correct ratio for the flag?
 - (a) 27 ft by 53 ft
 - (b) 35 ft by 66 ft
 - (c) 25 ft by 47 ft
 - (d) 30 ft by 53 ft
 - (e) 30 in. by 57 in.
5. Let $I=[0,1]\times[0,1]$. Suppose $E = (x,y) \in I : \sin(x) < \frac{1}{2}, \cos(x+y) \text{ is irrational}$, what is the Lebesgue measure of E ?
 - (a) 0.7853981634
 - (b) 1.0471975512

- (c) 1.8849555922
- (d) 0.6283185307
- (e) 0.2617993833

True / False (10 marks)

Answer True or False

6. True or False: The correct answer to 'Is the Taylor Series for f at $x=5$ where $f(x)=\sum_{n=0}^{\infty} \frac{x^n}{n!} ab$
6. True or False: The integral $\int_C \frac{dz}{z(z-2)^2}$ evaluates to 0.1234 when computed along the contour $C : |z - 2| = 1$.
7. True or False: A sample with a standard deviation of zero can still contain outliers that do not affect the sample's mean.
8. True or False: The function $f(x, y) = x^3 + y^3 + 3xy$ has local minima at both distinct points P and Q.
9. True or False: The probability that a random integer from the set S is divisible by 9 is 0.0455.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the waiting times for a new roller coaster ride, which are normally distributed with a mean of 35 minutes and a standard deviation of 10 minutes. Calculate and discuss the shortest time interval that accommodates 100,000 out of 150,000 riders, providing a thorough explanation of your methodology and findings.
12. Consider a point in space defined by coordinates (x, y, z) , constrained within the cube $-1 \leq x \leq 1, -1 \leq y \leq 1, -1 \leq z \leq 1$. Analyze the probability that this point lies within the unit sphere defined by the inequality $x^2 + y^2 + z^2 \leq 1$. Provide a detailed explanation of your reasoning and calculations.

End of Paper